MILITARY SPECIFICATION

RIOT CONTROL AGENT, CR SOLUTION

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a solution of Riot Control Agent, CR in propylene glycol and water.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 through 12 Gallons)

MILITARY

MIL-R-51446 - Riot Control Agent, CR

FSC 1365

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:00Z
Check the source to verify that this is the currentversion before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

2.2 Other Government documents, drawings, and pulbications. The following other Government documents, drawings, and pulbications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

18-60-1327 - Marking, Shipping Container, CR Solution
116-3-47 - Adapter Assembly

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D1193 - Reagent Water
E203 - Water Using Karl Fischer Reagent

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:00Z
Check the source to verify that this is the current version before use.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Composition. The composition of CR solution shall conform to table I when tested as specified therein. The CR solution shall be a thoroughly blended, homogeneous solution of the ingredients.

TABLE I. Composition

Ingredient	Percent by weight	Test paragraph
Riot Control Agent, CR conforming to MIL-R-51446	0.10 ± 0.01	4.2.3.1
Water conforming to ASTM D1193	20.0 ± 2.0	4.2.3.2
Propylene glycol, U.S.P.	Remainder	

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the CR solution produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). No lot of CR solution shall contain more than one lot of Riot Control Agent, CR.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:00Z
Check the source to verify that this is the current version before use.

4.2.2 Sampling and inspection procedure.

4.2.2.1 For examination of preparation for delivery. Sampling shall be conducted in accordance with MIL-STD-105 using an AQL of 2.5 percent defective. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers and the preparation for delivery thereof shall be examined for the following defects:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Tamper proof seal missing
(e) Closure torque not as specified
(f) Nozzle not crimped tight in pail head
(g) Container damaged or leaking
(h) Marking incorrect, missing, or illegible
(i) Palletization not as specified
(j) Container exterior contaminated with CR solution.

4.2.2.2 For CR solution test (see 6.7).

(a) CR in containers. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 20 milliliters (ml) shall be removed from each sample unit container and placed in a suitable clean, dry, closed container labeled to identify the lot and container from which it was taken. Each specimen shall be tested as specified in 4.2.3. Failure of any test by any specimen shall be cause for rejection of the lot represented.

TABLE II. Sampling for CR solution test

Number of containers in batch or lot	Number of sample containers
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

(b) CR in bulk quantities. Three specimens of approximately 20 ml each shall be removed from different parts of the lot and placed in suitable clean, dry, closed containers labeled to identify the lot from which it was teaken. Each specimen shall be tested as specified in 4.2.3. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.2.3 For container leakage test (see 6.7). Sampling shall be conducted in accordance with table II. Sample containers shall be tested for leakage as specified in 4.2.4. Evidence of leakage of any container shall be cause for rejection of the lot represented.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:00Z
Check the source to verify that this is the currentversion before use.

4.2.3 CR solution tests (see 6.7). Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.3.1 CR content.

(a) CR stock solution. Weigh to the nearest 0.1 milligram (mg) approximately 0.2 g of CR of known purity (see 6.6). Quantitatively transfer the CR to a 1-liter volumetric flask. Dissolve the CR and dilute to the volume mark with 1N hydrochloric acid. Make sure that the CR is completely dissolved and that the solution is thoroughly mixed.

(b) Preparation of calibration curve. Transfer aliquots of 2, 4, 6, 8, and 10 ml of the stock solution prepared in (a) into separate 100-ml volumetric flasks. Dilute each aliquot to the volume mark with 1N hydrochloric acid and mix thoroughly. Using a UV spectrophotometer suitable for measuring absorbance at 315 nanometers and cells of 1 centimeter path length, measure the absorbance of each of the diluted solutions at a peak occurring at 315 to 320 nanometers with 1N hydrochloric acid in the reference cell. Plot the absorbance of each diluted solution against its concentration in milligrams per liter using rectangular coordinate papar. Plot the standard curve using the method of least squares as the line determining procedure. Calculate the concentration of each diluted solution as follows:

Concentration of diluted solution in milligrams per liter = 10AB

where: A = Milliliters in transfered stock solution aliquot and
B = Concentration of stock solution in milligrams per milliliter.

(c) Procedure. Using a weighing buret, transfer 0.4 to 0.8 g of specimen to a 1-liter volumetric flask. Dilute to the volume mark with 1N hydrochloric acid, and mix thoroughly. Measure the absorbance of the solution at 315 to 320 nanometers using the same instrument and conditions as used in (b). Determine the concentration of CR in milligrams per liter from the calibration curve prepared in (b). Calculate the percent by weight CR in the specimen as follows:

$$\text{Percent CR} = \frac{AB}{1000W}$$

where: A = Concentration of CR in milligrams per liter from calibration curve,
B = Purity of CR standard, and
W = Weight of specimen in grams.

4.2.3.2 Water content. Determine the percent by weight water in the specimen in accordance with ASTM E203.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:00Z
Check the source to verify that this is the current version before use.

4.2.4 Container leakage. Invert the filled pail and place it in an inverted position for a minimum of 4 hours on a piece of paper. Examine the pail for evidence of leakage of contents during and at the end of the 4-hour period.

5. PACKAGING

5.1 Packing. Packing shall be level A or B as specified (see 6.2).

5.1.1 Level A. A quantity of 3 US gallons (+0 or -4 fluid ounces) of CR solution shall be packed in a clean, nominal 3-gallon capacity steel pail conforming to PPP-P-704, type I, class 4, except that the top cover shall be minimum 22 gage. The pail shall be furnished with a screw-cap closure and a commercial nominal 1-3/4-inch clinched-in nozzle without push-pull or equivalent pouring spout. The nozzle shall be compatible with the female end of the adapter assembly shown in Drawing 116-3-47. The facing on the liner for the screw cap shall be highly resistant to propylene glycol-water solutions. A metal tamper-proof inner seal shall also be furnished with each pail. The entire inner surface of each pail shall have two coats of clear baked-on epoxy-phenolic resin. The cap shall be screwed on the nozzle of the pail to a torque in accordance with the pail supplier's recommendation (see 6.5) for closure of the filled pail after inserting inner seal, in accordance with the appendix to PPP-P-704. The pail shall be free of CR solution contamination after filling and closing. After filling and closing, containers shall not leak when tested in accordance with 4.2.4.

5.1.2 Level B. A quantity of 55 gallons ±1 quart of CR solution shall be packed in a nominal 55-gallon capacity steel drum conforming to Department of Transportation (DOT) Specification 5B. The entire interior surface of the drum shall be protected by two coats of clear baked-on epoxy-phenolic resin. The 3/4-inch and 2-inch openings shall be in the top head of the drum, and their threaded closures shall be protected by tamper-proof outer seals.

5.2 Marking. Pails shall be marked as shown on Drawing 18-60-1327. Drums and pallet loads shall be marked in accordance with DOT regulations and MIL-STD-129.

5.3 Unitization. Forty-eight pails shall be palletized per pallet in accordance with MIL-STD-147, load type III.

6. NOTES

6.1 Intended use. CR solution is intended for use in the M33A1 and M36 Riot Control Agent Dispensers.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packing (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:00Z
Check the source to verify that this is the currentversion before use.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

6.5 Ordering of empty pails. When ordering new empty pails, the contractor must obtain torque closure data from the pail supplier.

6.6 Standard CR. CR of known purity for use in 4.2.3.1(a) may be obtained from Commander, US Army Armament Research and Development Command, Product Assurance Directorate, Chemical Systems Division, Aberdeen Proving Ground, MD 21010. Alternatively, the contractor may prepare his own standard CR and obtain verification of its purity from Chemical Systems Division.

6.7 Sampling and testing precautions. This specification covers inspection of chemical material which is potentially hazardous to personnel. All applicable safety rules, regulations and procedures must be followed in the sampling and testing of these materials.

6.8 Supersession data. This specification supersedes and includes the requirements of Edgewood Arsenal Purchase Description EA-R-1136, dated 5 January 1976.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1365-A072

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:00Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
NALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armament Research & Development
Command
Attn: DRDAR-TSC-S
Aberdeen Proving Ground, MD 21010

OLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-R-51447(EA) Riot Control Agent, CR Solution

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:00Z
Check the source to verify that this is the current version before use.